Girls Build Bots — Junior Coding Basics

Badge-aligned workshop guide for robotics-team-led outreach

Level: Juniors (Grades 4–5) · **Badge:** Coding for Good — Coding Basics **Series:** Coding for Good (Coding Basics · Digital Game Design · App Development) **Format:** One 90–120 minute session · **Ratio:** 1 mentor per 4–5 Scouts **Who runs it:** A FIRST robotics team (FLL / FTC / FRC) leading the session. A physical robot is a nice bonus but **not required** — every activity runs on free browser tools the team or troop can provide.

> **Troop Leaders:** This curriculum is designed to *align with* the Junior Coding for Good badge series. Official badge requirements are in your Girl Scout Volunteer Toolkit (VTK). Verify completed activities against the official steps before awarding badges. Badges are purchased through your council shop.

The through-line: coding *for good*

Every badge in this series ties coding to helping people. Juniors are ready to hear it plainly: programmers build things that solve problems for real users. Mentors: name the "who is this for" in every activity.

Vocabulary: algorithm, program, sequence, loop, event, bug, debug, pseudocode.

Coding Basics (90–120 minutes)

Badge theme: Understanding how coders break big problems into steps a computer can follow — algorithms, sequences, loops, and events.

Materials

- Laptops with **MakeCode Arcade** (arcade.makecode.com) or **Scratch** (scratch.mit.edu)
- Index cards for the unplugged algorithm relay
- Projector for live coding

Activity 1 — Algorithm Relay (15 min)

Teams write an algorithm on cards for a physical task (stack three cups into a tower, do a 4-move dance). Another team executes it *exactly as written* — no interpreting! Missing steps and wrong order surface instantly. **Debrief:** computers do exactly what we say, so precision is the coder's job.

Activity 2 — Loops Save Work (15 min)

Rewrite a repetitive algorithm ("clap, clap, clap, clap, clap, clap") using a **loop** ("repeat 6: clap"). Then find the loop hiding in real life: brushing every tooth, dealing every card. Loops are how coders avoid writing the same steps forever.

Activity 3 — Build an Interactive Program (50 min)

Mentor-led progressive build in MakeCode Arcade, one concept per stage: 1. **Sequence** — a sprite appears and walks across the screen 2. **Event** — the sprite moves when arrow keys are pressed (*when this happens, do that*) 3. **Loop** — a star respawns *forever* after being collected 4. **Challenge** — add a counter that tracks how many stars you've collected

Fast finishers become deputy mentors for a neighbor — teaching is the best proof of understanding.

Activity 4 — Debug Hunt (15 min) + Wrap (10 min)

Mentors hand each pair a short broken program (prepared in advance: a wrong event, a loop that never runs). Scouts find and fix the bug, then explain what was wrong. Closing circle: everyone names the concept they'd teach a friend first — sequence, loop, or event.

> **Robot moment (optional):** Share for 60 seconds how the team's robot code uses the exact same three ideas — sequences for autonomous routines, events for driver buttons, loops for checking sensors — just in a bigger language.

Completion Checklist — Coding Basics

By the end of the session, to earn the badge, each Scout can: - [ ] Write a precise algorithm another person (or computer) can follow exactly - [ ] Explain what a loop does and why coders use them - [ ] Build a program that uses a sequence, an event, and a loop - [ ] Find and fix a bug in a program and explain what was wrong

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Mentor Guide (for the robotics team)

- **You don't need a robot for this.** Free browser tools carry the whole session. The "robot moment" box is a 60-second add-on.
- **Let the relay fail funny.** Execute bad algorithms literally and cheerfully — the laughter is the lesson sticking.
- **Ask, don't tell.** Juniors can reason: "what do you think happens if we move this block?"
- **Let pairs struggle for three minutes** before stepping in — productive struggle is where learning happens.
- **Deputy mentors are a feature.** Promote fast finishers to helpers; it keeps the room moving and builds their confidence.

📣 Share Your Impact — required for teams and troops using this curriculum

If you use Girls Build Bots materials to run a workshop, please complete all three:

1. **Credit us.** Tell your Scouts and audience that the curriculum comes from Girls Build Bots, and credit "Girls Build Bots" anywhere you share about the event.
2. **Tag us.** Post about your workshop and tag **@girlsbuildbots** on Instagram.
3. **Report your results.** Email **girlsbuildbots@gmail.com** with: the number of girls who participated, the number of badges earned, and a sentence or two of feedback on how it went.

This is how the program grows and how we measure our impact — your report becomes part of the Girls Build Bots story and helps the next troop and team.

Created for the "Girls Build Bots" Girl Scout Gold Award project. Original curriculum aligned to the Junior Coding for Good badge series; official requirements © Girl Scouts of the USA, available to troop leaders in the Volunteer Toolkit.